

Mandatory Assignment 2

Applied Statistics, IT University of Copenhagen

Matus Stanko

Case 4.

A classical dataset contains data on body weight and heart weight for 97 male cats and 47 female cats (Fisher, 1947). The dataset `cats` is available in the `MASS` package for R. It contains the variables `Sex` (sex of the cat), `Bwt` (body weight in kilograms), and `Hwt` (heart weight in grams).

1.

Use the following commands to make and attach a dataset, `males`, with the data for male cats only:

```
library(MASS)
males <- subset(cats, Sex=="M") # select from cats only where Sex is M (Male)
attach(males)
```

2.

Run the command:

```
summary(lm(Hwt~Bwt))

##
## Call:
## lm(formula = Hwt ~ Bwt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.7728 -1.0478 -0.2976  0.9835  4.8646
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.1841     0.9983  -1.186   0.239
## Bwt           4.3127     0.3399  12.688 <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.557 on 95 degrees of freedom
## Multiple R-squared:  0.6289, Adjusted R-squared:  0.625
## F-statistic:   161 on 1 and 95 DF,  p-value: < 2.2e-16
```

It seems like linear regression model, the dependent variable (Hwt/Bwt) is regressed on a single predictor variable (Bwt):

$$\beta_0 + \beta_1 \cdot Bwt + \epsilon$$

Where: β_0 is intercept coefficient = -1.1841

β_1 is coefficient for variable Bwt = 4.3127

ϵ is error

the slope parameter of 4.3127 means that for every unit increase in bodyweight the expected ratio of heartweight to bodyweight increases by approximately 4.3127 units. The slope is positive, so line representing expected ratios will be growing (the higher the body weight, the higher the expected heart weight)

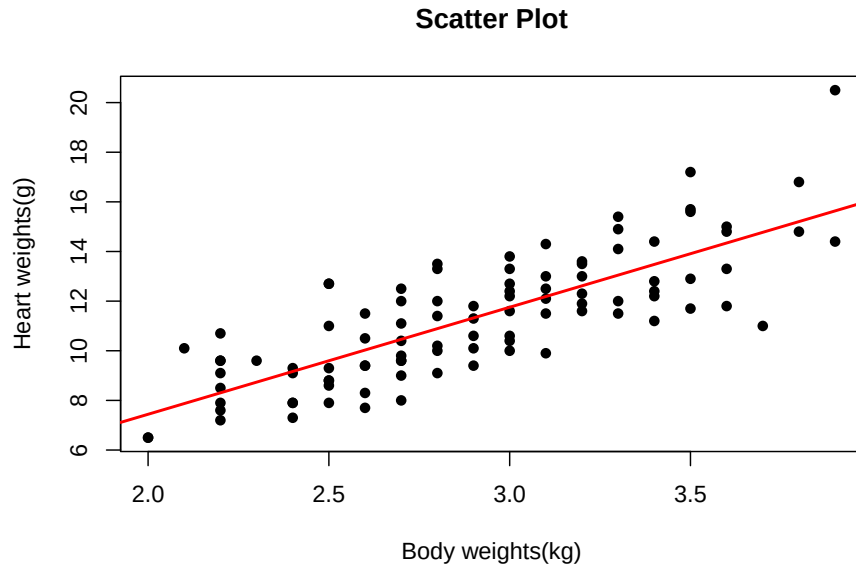
3.

Make a scatter plot of the data and add the estimated regression line with the `abline()` function

```
#Fit lin. regression
fit <- lm(Hwt ~ Bwt)

#Create scatter plot graph
plot(Bwt, Hwt,
     main = "Scatter Plot",
     xlab = "Body weights(kg)",
     ylab = "Heart weights(g)",
     col = "black",
     pch = 16)

#Draw regression line
abline(fit, col = "red", lwd = 2)
```



4.

Compute the 95% confidence interval for the intercept parameter and for the slope parameter. Consider intercept parameter as 0.

$$y_i = \beta \cdot x_i + \epsilon_i$$

```
confint(fit)
```

```
##                2.5 %    97.5 %
## (Intercept) -3.165939 0.7977636
## Bwt         3.637904 4.9874540
```

This command will give us 2 confidence intervals, first row (intercept) is calculating confidence interval without intercept $[-3.16; 0.79]$ and $[3.63; 4.98]$ meaning that I am 95% sure that true value of these parameters are in these intervals.

While considering intercept (intercept = 0), the model will look like this:

$$y_i = \beta \cdot x_i + e_i$$

This could be better fit for our data, since logically if body weight is 0, then heart weight must be 0. So this could better approx. the regression model

5.

Use one of the confidence intervals from question 4 to explain why this is a reasonable model for the cat data.

Lets take option with intercept = 0. This means that regression line will be crossing the origin[0,0].

The regression model (line) will tell me that: If bodyweight is 0, then also heartweight is 0. This makes sense. The confidence interval without intercept [-3.16;0.79] includes number 0 so it is possible that true parameter is 0, however It does not have to be. But In this case it make sense to use intercept = 0.

Also true value of slope is possitive (based on 95% CI), I know that regression line is growing, meaning that: heartweight is growing with bodyweight.

6.

In order to derive the least squares estimate of β , the function:

$$Q(\beta) = \sum_{i=1}^n (y_i - \beta \cdot x_i)^2$$

should be minimalized. Show that $Q(\beta)$ is the smallest possible for :

$$\beta = \hat{\beta} = \frac{\sum_{i=1}^n x_i \cdot y_i}{\sum_{i=1}^n x_i^2}$$

$$Q(\beta) = \sum_{i=1}^n (y_i - \beta \cdot x_i)^2$$

Solve for Beta: Derivate with respect to β

$$\frac{\partial}{\partial \beta} Q(\beta) = \sum_{i=1}^n 2 \cdot (y_i - \beta \cdot x_i) \cdot \frac{\partial}{\partial \beta} (y_i - \beta \cdot x_i)$$

$$\frac{\partial}{\partial \beta} Q(\beta) = \sum_{i=1}^n 2 \cdot (y_i - \beta \cdot x_i) \cdot (-x_i) = -2 \sum_{i=1}^n (y_i - \beta \cdot x_i) \cdot x_i$$

Now set it to 0

$$-2 \sum_{i=1}^n y_i \cdot x_i - \beta \sum_{i=1}^n x_i^2 = 0 \text{ multiply by } -\frac{1}{2}$$

$$\sum_{i=1}^n y_i \cdot x_i - \beta \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n y_i \cdot x_i = \beta \sum_{i=1}^n x_i^2$$

$$\beta = \frac{\sum_{i=1}^n y_i \cdot x_i}{\sum_{i=1}^n x_i^2} = \hat{\beta}$$

7. Compute $\hat{\beta}$ for the cat data (use the sum() function).

```
#For male cats
numerator <- sum(Bwt * Hwt)
denominator <- sum(Bwt^2)
beta_hat <- numerator / denominator

print(beta_hat)
```

```
## [1] 3.91461
```

```
#For all cats
numerator_all <- sum(cats$Bwt * cats$Hwt)
denominator_all <- sum(Bwt^2)
beta_hat_all <- numerator / denominator
```

8.

Run the following command and recognize the value of $\hat{\beta}$ in the output:

```
summary(lm(Hwt~Bwt-1))
```

```
##
## Call:
## lm(formula = Hwt ~ Bwt - 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4841 -1.1097 -0.2268  0.9879  5.2330
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## Bwt   3.91461    0.05393   72.58  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.56 on 96 degrees of freedom
## Multiple R-squared:  0.9821, Adjusted R-squared:  0.9819
## F-statistic: 5268 on 1 and 96 DF, p-value: < 2.2e-16
```

I can see estimate of β 3.91461, that is the same as my calculations using sum

9.

Add the new fitted regression line to the plot from question 3. You may use another color or another line type; try, for example, one of the options `col="red"` or `lty=2` to `abline()`.

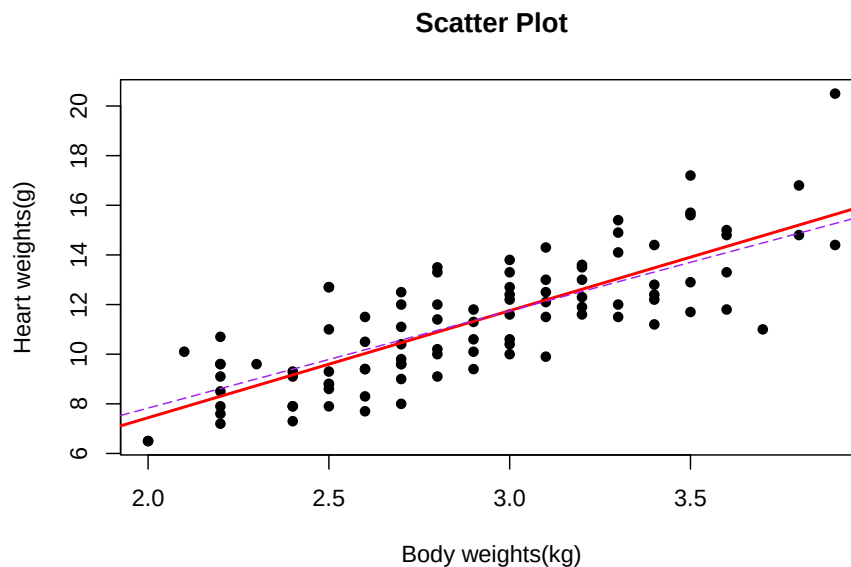
```

fit_no_intercept <- lm(Hwt~Bwt-1) #-1 means no intercept

#Create scatter plot graph
plot(Bwt, Hwt,
     main = "Scatter Plot",
     xlab = "Body weights(kg)",
     ylab = "Heart weights(g)",
     col = "black",
     pch = 16)

#Regression line n.1
abline(fit, col = "red", lwd = 2)
#Regression line n.2
abline(fit_no_intercept, col = "purple", lty = 5)

```



10.

Compute an estimate for the expected heart weight for a random cat with body weight 2.5 kilograms.

```

body_weight <- 2.5

# Expected heart weight using regression model without intercept
expected_heart_weight <- beta_hat * body_weight

print(expected_heart_weight)

```

```
## [1] 9.786525
```

So expected heart weight for cat with Bwt of 2.5kg, is 9.78g

11.

Find $SE(\hat{\beta})$ in the output from question 8 and use it to compute the 95% confidence interval for β without the use of the `confint()` function.

$$CI = \hat{\beta} \pm t_{0.975, n-p} \cdot SE(\hat{\beta})$$

Where:

$\hat{\beta}$ is β estimate,

t is critical value,

p is the number of predictors (in this case only slope = 1)

```
#variables known for male cats
SE = 0.05393
n = 97 #or 2 <- length(Hwt)

#Degree of freedom
## formula is n-p. N = 97 and p = 1 the number of predictors. Since I got SE from lm(formula)
degree_of_freedom = n-1

alpha <- 0.05

t_critical <- abs(qt(1 - alpha / 2, degree_of_freedom))

#Using formula
low_CI = beta_hat - t_critical * SE
upp_CI = beta_hat + t_critical * SE

print(paste("CI is: [", low_CI, "; ", upp_CI, "]"))
```

```
## [1] "CI is: [ 3.80755989467535 ; 4.02166030251616 ]"
```

So true value of β with SE=0.05393 will be in CI with 95%

12.

Use `confint()` to compute the same confidence intervals. Compare it to the confidence interval for the slope parameter from question 4.

```
fit <- lm(Hwt ~ Bwt - 1)
conf_interval <- confint(fit, level = 0.95)

# Print the confidence intervals
print(conf_interval)
```

```
##          2.5 %    97.5 %  
## Bwt 3.807551 4.021669
```

Confidence interval got in question 12 [3.80;4.02] is more narrow what interval in question 4 [3.63;4.98], in question 4 I did conf. intervals with 2 parameters - slope and intercept, however in question 12 only slope. This brings additional uncertainty because I am estimating 2 parameters that made confidence interval wider.

13.

```
t.test(Hwt/Bwt)
```

```
##  
## One Sample t-test  
##  
## data: Hwt/Bwt  
## t = 71.743, df = 96, p-value < 2.2e-16  
## alternative hypothesis: true mean is not equal to 0  
## 95 percent confidence interval:  
## 3.786792 4.002302  
## sample estimates:  
## mean of x  
## 3.894547
```

Statistical model is One Sample t-test (determines whether the mean of a sample differs significantly from a specific value), we are testing the ratio of heart weight to body weight of cats.

The estimated mean $\bar{x}$ represents mean value of this sample data. So Hwt/Bwt

Difference between one sample t-test and linear regression model without intercept: a.) Linear regression model without intercept: Fits a regression line through origin [0,0]. Coefficients represent the slope of the line, so how much response variable changes for change in predictor variable.

b.) Compares the mean of sample to population mean, and deciding if the sample mean differs (significantly) from population mean. Using t-value and p-value.

This mandatory assignment helped me understand when and why not to use an intercept in linear regression models. By not using the intercept, I gained insights into obtaining more precise confidence intervals, which are showing where the true parameter is most likely to lie.